

Lecture 16 – Gauss's Law and Applications

- **A/Prof Tay Seng Chuan**
Email: pvotaysc@nus.edu.sg

Have you ever thought of these questions?

- how you can determine the amount of charge within a closed surface by examining the electric field on the surface.
- what is meant by electric flux, and how to calculate it.
- how Gauss's law relates the electric flux through a closed surface to the charge enclosed by the surface.
- how to use Gauss's law to calculate the electric field due to a symmetric charge distribution.
- where the charge is located on a charged conductor.



1

A few slides in this chapter come from Chapter-22 of Young's textbook.

Introduction

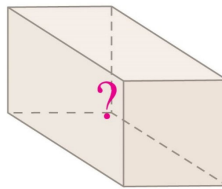
- This child acquires an electric charge by touching the charged metal shell.
- The charged hairs on the child's head repel and stand out.
- What would happen if the child stood **inside** a large, charged metal shell?
- Symmetry properties play an important role in physics.
- Gauss's law will allow us to do electric-field calculations using symmetry principles.



2

What Is Gauss's Law All About?

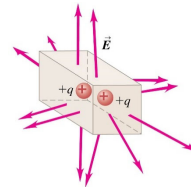
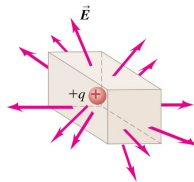
- Given any general distribution of charge, we surround it with an imaginary surface that encloses the charge.
- Then we look at the electric field at various points on this imaginary surface.
- Gauss's law is a relationship between the field at all the points on the surface and the total charge enclosed within the surface.



3

Charge and Electric Flux (1 of 2)

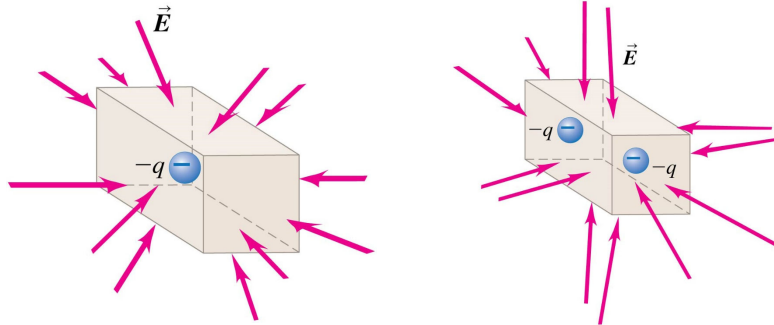
- In both boxes below, there is a **positive** charge within the box on the left, and two positive point charges on the right, and the **electric flux** is produced **outward** pointing through the surface of the box.
- The field patterns on the six surfaces of the boxes are different in detail, since the box on the left contains one point charge, and the box on the right contains two.



4

Charge and Electric Flux (2 of 2)

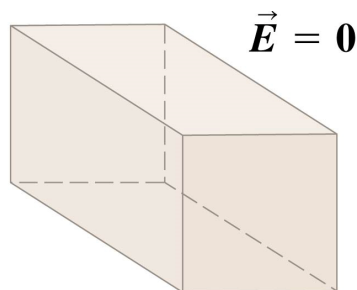
- When there are negative charges inside the box, there is an **inward** pointing electric flux on the surface.



5

Zero Net Charge Inside a Box (1 of 3)

- What happens if there is zero charge inside the box?
- If the box is empty and the electric field is zero everywhere, then there is no electric flux into or out of the box.

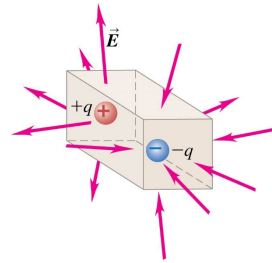


6

Zero Net Charge Inside a Box (2 of 3)

- What happens if there is zero **net** charge inside the box?
- There is an electric field, but it “flows into” the box on half of its surface and “flows out of” the box on the other half.

- Hence there is no **net** electric flux into or out of the box.

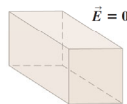


7

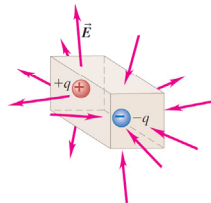
Zero Net Charge Inside a Box (3 of 3)

- What happens if there is charge **near** the box, but not inside it?
- On one end of the box, the flux points into the box; on the opposite end, the flux points out of the box; and on the sides, the field is parallel to the surface and so the flux is zero.
- The **net** electric flux through the box is zero.

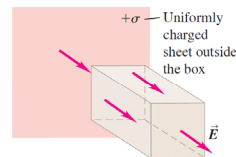
(a) No charge inside box,
zero flux



(b) Zero *net* charge inside box,
inward flux cancels outward flux.



(c) No charge inside box,
inward flux cancels outward flux.

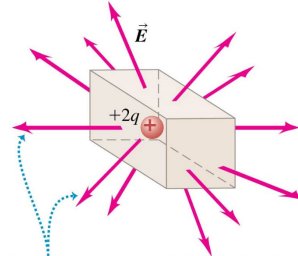
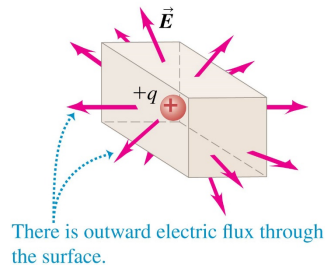


8

What Affects the Flux Through a Box? (1 of 2)

- The net electric flux is **directly proportional** to the net amount of charge enclosed within the surface.

(a) A box containing a positive point charge $+q$



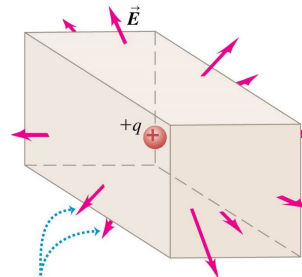
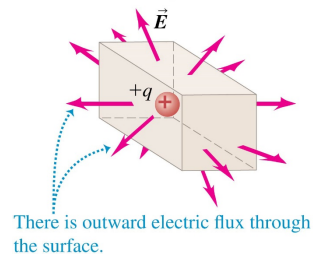
Doubling the enclosed charge also doubles the magnitude of the electric field on the surface, so the electric flux through the surface is twice as great as in (a).

9

What Affects the Flux Through a Box? (2 of 2)

- The net electric flux is **independent** of the size of the closed surface.

(a) A box containing a positive point charge $+q$



Surface is larger but the electric field on the surface is weaker!!

The electric flux is the same as in (a): The magnitude of the electric field on the surface is $\frac{1}{4}$ as great as in (a), but the area through which the field "flows" is 4 times greater.

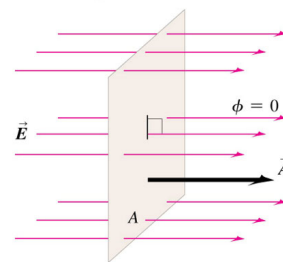
10

Calculating Electric Flux (1 of 3)

- Consider a flat area perpendicular to a uniform electric field.
- Increasing the area means that more electric field lines pass through the area, increasing the flux.
- A stronger field means more closely spaced lines, and therefore more flux.

Surface is face-on to electric field:

- $\vec{E}$ and $\vec{A}$ are parallel (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 0$).
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA$.



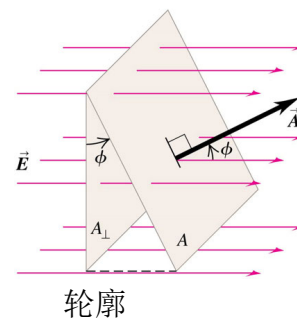
11

Calculating Electric Flux (2 of 3)

- If the area is not perpendicular to the field, then fewer field lines pass through it.
- In this case the area that counts is the silhouette area that we see when looking in the direction of the field.

Surface is tilted from a face-on orientation by an angle ϕ :

- The angle between $\vec{E}$ and $\vec{A}$ is ϕ .
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos \phi$.

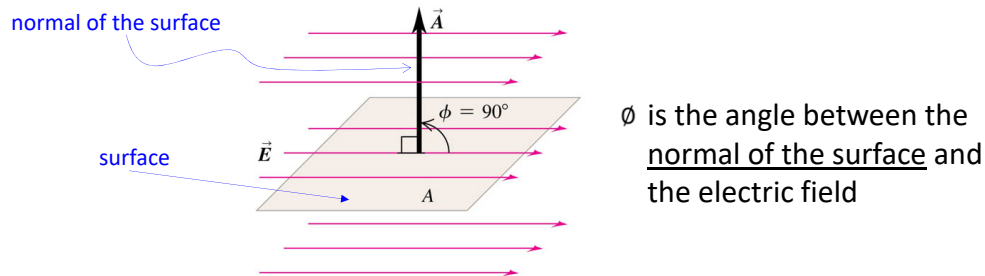


12

Calculating Electric Flux (3 of 3)

- If the area is edge-on to the field, then the area is perpendicular to the field and the flux is zero.

- Surface is edge-on to electric field:
- $\vec{E}$ and $\vec{A}$ are perpendicular (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 90^\circ$).
 - The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos 90^\circ = 0$.



13

Flux of a Nonuniform Electric Field

- In general, the flux through a surface must be computed using a **surface integral** over the area:

$$\Phi_E = \int E \cos \phi \, dA = \int E_{\perp} \, dA = \int \vec{E} \cdot d\vec{A}$$

Electric flux through a surface

Magnitude of electric field E

Component of $\vec{E}$ perpendicular to surface

Angle between $\vec{E}$ and normal to surface

Element of surface area

Vector element of surface area

- The SI unit for electric flux is $1 \text{ N} \cdot \frac{\text{m}^2}{\text{C}}$.

14

Gauss's Law

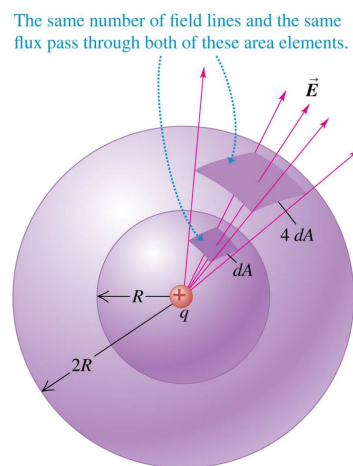
- Carl Friedrich Gauss helped develop several branches of mathematics, including differential geometry, real analysis, and number theory.
- The “bell curve” of statistics is one of his inventions
- Gauss also made state-of-the-art investigations of the earth’s magnetism and calculated the orbit of the first asteroid to be discovered.
- While completely equivalent to Coulomb’s law, **Gauss’s law** provides a different way to express the relationship between electric charge and electric field.



15

Point Charge Centered in a Spherical Surface

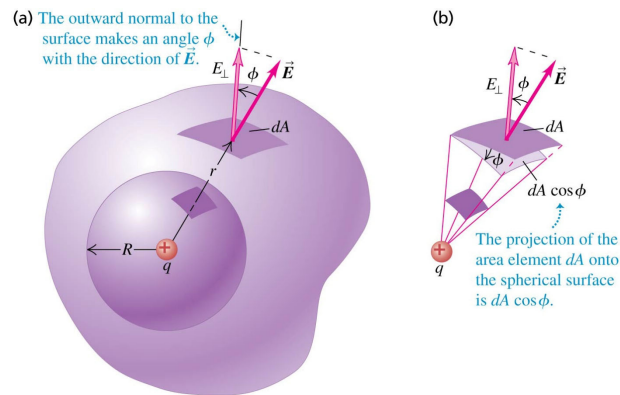
- The figure shows the projection of an element of area dA of a sphere of radius R onto a concentric sphere of radius $2R$.
- The area element on the larger sphere is $4 dA$, but the electric field magnitude is $\frac{1}{4}$ as great on the sphere of radius $2R$ as on the sphere of radius R .
- Hence the electric flux is the **same** for both areas and is **independent** of the radius of the sphere.



16

Point Charge Inside a Nonspherical Surface

- As before, the flux is independent of the surface and depends only on the charge inside.



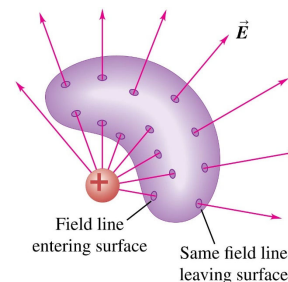
17

Gauss's Law in a Vacuum

- For a closed surface **enclosing no charge**:

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = 0$$

- The figure shows a point charge **outside** a closed surface that encloses no charge.
- If an electric field line from the external charge enters the surface at one point, it must leave at another.



18

General Form of Gauss's Law

- Let Q_{encl} be the total charge enclosed by a surface.
- Gauss's law states that **the total electric flux through a closed surface is equal to the total (net) electric charge inside the surface, divided by ϵ_0** :

Gauss's law:

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

Electric flux through a closed surface of area A = surface integral of $\vec{E}$

Total charge enclosed by surface

Electric constant

19

Positive and Negative Flux

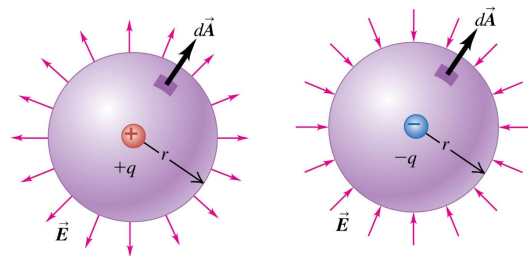
- In the case of a single point charge at the center of a spherical surface, Gauss's law follows simply from Coulomb's law.

$$\phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

If the enclosed sphere have the radius r ,

$$E \times (4\pi r^2) = \frac{Q}{\epsilon_0} \rightarrow E = \frac{Q}{4\pi r^2 \epsilon_0}$$

$$= k \frac{Q}{r^2} \text{ (same answer!!)}$$



- A surface around a positive charge has a positive (outward) flux, and a surface around a negative charge has a negative (inward) flux.

20

Applications of Gauss's Law

(1 of 2)

Gauss's law: $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$

Electric flux through a closed surface of area A = surface integral of $\vec{E}$

Total charge enclosed by surface

Electric constant

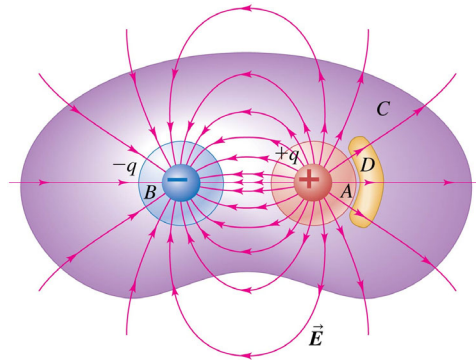
- Without having to do any integration, we can use Gauss's law to determine the electric flux through the closed surfaces in the diagram.

- $\Phi_{EA} = +\frac{q}{\epsilon_0}$

- $\Phi_{EB} = -\frac{q}{\epsilon_0}$

- $\Phi_{EC} = 0$

- $\Phi_{ED} = 0$



21

Applications of Gauss's Law

(2 of 2)

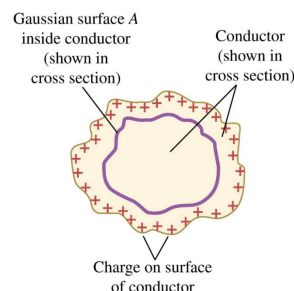
Gauss's law: $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$

Electric flux through a closed surface of area A = surface integral of $\vec{E}$

Total charge enclosed by surface

Electric constant

- Suppose we construct a Gaussian surface inside a conductor.
- Because $\vec{E} = 0$ everywhere on this surface, Gauss's Law requires that the net charge **inside** the surface is zero.
- Under electrostatic conditions (charges not in motion), any excess charge on a solid conductor resides entirely on the conductor's surface.



22

Field of a Uniform Line Charge

Gauss's law: $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$

Electric flux through a closed surface of area A = surface integral of $\vec{E}$

Total charge enclosed by surface

Electric constant

- Electric charge is distributed uniformly along an infinitely long, thin wire. The charge per unit length is λ (assumed positive).

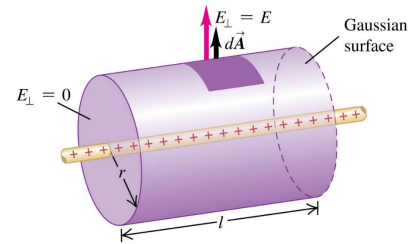
- Using Gauss's law, we can find the electric field for the uniform line charge as $E = \frac{1}{2\pi\epsilon_0} \left(\frac{\lambda}{r} \right)$

$$\phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

If the enclosed cylinder has the radius r , and height ℓ , for the electric flux on the lateral surface of the cylinder,

$$E \times (2\pi r \times \ell) = \frac{\lambda \ell}{\epsilon_0} \rightarrow E = \frac{1}{2\pi\epsilon_0} \left(\frac{\lambda}{r} \right)$$

because the electric flux on the top and bottom circular area (left and right in this diagram) is 0 (no penetration of electric field)



23

Field of a Uniform Line Charge

Gauss's law: $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$

Electric flux through a closed surface of area A = surface integral of $\vec{E}$

Total charge enclosed by surface

Electric constant

- Electric charge is distributed uniformly along an infinitely long, thin wire. The charge per unit length is λ (assumed positive).

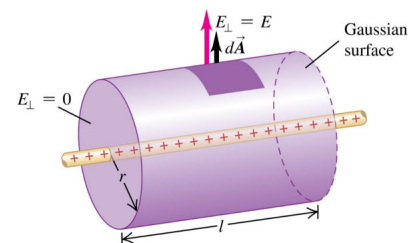
- Using Gauss's law, we can find the electric field for the uniform line charge as $E = \frac{1}{2\pi\epsilon_0} \left(\frac{\lambda}{r} \right)$

$$\phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

If the enclosed cylinder has the radius r , and height ℓ , for the electric flux on the lateral surface of the cylinder,

$$E \times (2\pi r \times \ell) = \frac{\lambda \ell}{\epsilon_0} \rightarrow E = \frac{1}{2\pi\epsilon_0} \left(\frac{\lambda}{r} \right)$$

because the electric flux on the top and bottom circular area (left and right in this diagram) is 0 (no penetration of electric field)



23

Field of an Infinite Plane Sheet of Charge

Gauss's law: $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$

Electric flux through a closed surface of area A = surface integral of $\vec{E}$

Total charge enclosed by surface

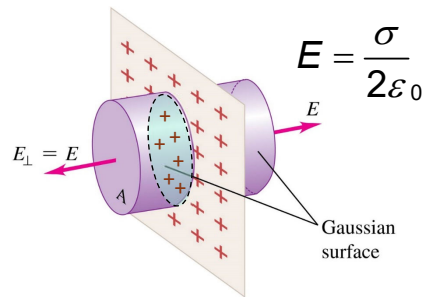
Electric constant

- Gauss's law can be used to find the electric field caused by a thin, flat, infinite sheet with a uniform positive surface charge density σ .

$$\phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

We only work the charges enclosed in a circle at the center of the sheet. Now we use a cylinder to enclosed the charges. If the top surface of the cylinder has an area A ,

$$\underbrace{E \times A}_{\text{left circle}} + \underbrace{E \times A}_{\text{right circle}} = \frac{\sigma A}{\epsilon_0} \rightarrow E = \frac{\sigma}{2\epsilon_0}$$



because the electric flux on the lateral surface of the cylinder is 0 (no penetration of electric field).

25

Field of an Infinite Plane Sheet of Charge

Gauss's law: $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$

Electric flux through a closed surface of area A = surface integral of $\vec{E}$

Total charge enclosed by surface

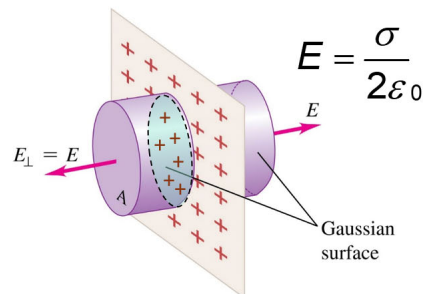
Electric constant

- Gauss's law can be used to find the electric field caused by a thin, flat, infinite sheet with a uniform positive surface charge density σ .

$$\phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

We only work the charges enclosed in a circle at the center of the sheet. Now we use a cylinder to enclosed the charges. If the top surface of the cylinder has an area A ,

$$\underbrace{E \times A}_{\text{left circle}} + \underbrace{E \times A}_{\text{right circle}} = \frac{\sigma A}{\epsilon_0} \rightarrow E = \frac{\sigma}{2\epsilon_0}$$



because the electric flux on the lateral surface of the cylinder is 0 (no penetration of electric field).

25

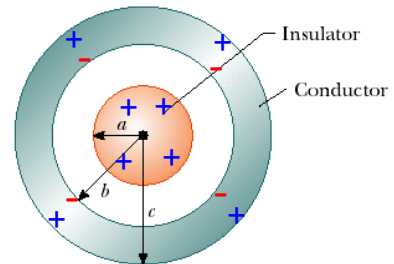
Example. A solid, insulating sphere of radius a has a uniform charge density ρ and a total charge Q . Concentric with this sphere is an uncharged, conducting hollow sphere whose inner and outer radii are b and c , as shown in Figure. (i) Find the magnitude of the electric field in the regions $r < a$, $a < r < b$, $b < r < c$, and $r > c$. (ii) Determine the induced charge per unit area on the inner and outer surfaces of the conducting hollow sphere.

Gauss's law: $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$

Electric flux through a closed surface of area A = surface integral of $\vec{E}$

Total charge enclosed by surface

Electric constant



For simplicity in drawing, let $Q = +4$

Answer:

$$(i) \ r < a: \ E_{r < a} \times (4\pi r^2) = \frac{\rho \times \left(\frac{4}{3}\pi r^3\right)}{\epsilon_0} \rightarrow E_{r < a} = \frac{\rho r}{3\epsilon_0}$$

$$a < r < b: \ E_{a < r < b} = \frac{Q}{4\pi r^2 \epsilon_0} \text{ since the total charge of the enclosure is } Q.$$

$$b < r < c: \ E_{b < r < c} = 0 \text{ since the total charge of the enclosure is } 0.$$

$$r > c: \ E_{r > c} = \frac{Q}{4\pi r^2 \epsilon_0} \text{ since the total charge of the enclosure is } Q.$$

$$(ii) \ \sigma_{\text{inner}} = \frac{Q}{4\pi b^2}, \ \sigma_{\text{outer}} = \frac{Q}{4\pi c^2}$$

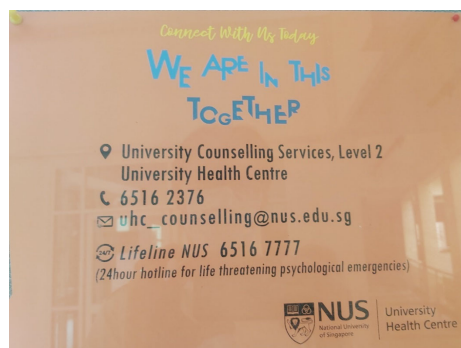
27

Helplines to de-stress

We are not superman/supergirl. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019

28